NGUYEN DUY TRANG

5+ Years Exp • Open to Opportunities

Mobile Engineer (Flutter, iOS, Android • Clean Architecture)

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PROFESSIONAL SUMMARY

OVERVIEW

Mobile Engineer with 5+ years of hands-on experience developing multi-platform and native applications (Flutter, iOS, Android, Web/Kiosk). Expert in standardized architecture (Clean Architecture, MVVM, Modular/Package-based), mastering both BLoC (flutter_bloc) and Riverpod with Unidirectional Data Flow, packaging Core Design Systems & Base Components (Custom Widgets, dynamic Theme Engine, Responsive Multi-screen Layout, Localization, Base Dio Client with Interceptors/Auto-Refresh Token/Retry Policy, Global Exception Handling). Deep understanding of Widget lifecycle and the Render Tree (Widget/Element/RenderObject), utilizing Flutter DevTools (Profiler, Memory, Network) to ensure smooth 60 FPS, zero jank/drop frames, and zero memory leaks. Specialized in Flutter Web optimization (GoRouter/AutoRoute nested routes, query params, CanvasKit/WASM, asset caching). Proven track record operating 1M+ download apps, establishing automated CI/CD pipelines (Fastlane, GitHub Actions, GitLab CI/CD, Docker), code review standards, and junior developer mentoring.

VERIFIED PRODUCTION HIGHLIGHTS

METRICS

- **1M+ Downloads:** Operating large-scale social travel network & real-time OTT on iOS, Android & Flutter.
- **99.5%+ Crash-Free:** Render Tree & DevTools Profiler optimization, locked 60 FPS & 0 memory leaks.
- **BLoC & Offline-First:** Unidirectional Data Flow, WebSocket real-time, SQLite/Drift & Hive caching.
- **Mobile & Web:** Modular Clean Architecture, responsive multi-screen layouts with GoRouter/AutoRoute.
- **Design System & Base:** Packaged Core UI Library, Theme Engine, Base Dio Client & Token Refresh.
- **277 Tests & CI/CD:** 100% pass rate (Unit/Widget/E2E), Fastlane, GitHub Actions & GitLab CI/CD.

TECHNICAL SKILLS MATRIX

COMPETENCIES

Languages & Core:	Flutter & Dart (Expert), Swift, SwiftUI, UIKit, Kotlin, HTML5/CSS3, TypeScript.
Architecture & State:	Clean Architecture (Package-based), BLoC (flutter_bloc), Riverpod 2.x, MVVM, Unidirectional Data Flow, SOLID, Modularization.
Design System & UI:	Core UI Library, Custom Widgets, Theme Engine (Dark/Light), Responsive Layout (Mobile/Web), Localization (intl).
Networking & Local DB:	Base Dio Client (Interceptors, Auto Refresh Token, Retry Policy), WebSocket/Socket.IO, SQLite/Drift, Hive, SharedPreferences/Isar.
Performance & Web:	Render Tree Optimization (RepaintBoundary, const), Flutter DevTools, GoRouter & AutoRoute (Nested Routes, URL Query Params), CanvasKit/WASM.
DevOps, QA & Mentoring:	Fastlane, GitHub Actions, GitLab CI/CD, Docker, 277 Automated Tests (100% Pass), PR Review, Coding Conventions & Mentoring.

FEATURED FLAGSHIP PROJECT

PERSONAL ARCHITECTURE

ASTROBITE — AI Food Scanner & Voice Calorie Tracker / AstroVoice AI

2026 - Present

Mobile Engineer & System Architect — Cross-Platform AI Mobile & Web Ecosystem

Tech Stack: Flutter 3.x, Dart 3.x, Clean Architecture (12 Modular Packages), Riverpod 2.6 (AsyncNotifier, code-gen), AutoRoute & GoRouter (Deep Linking, Web Query Params), Gemini 3.8 Flash (Vision & NLU), Base Dio Client, Custom Design System, Flutter DevTools (60 FPS, 0 memory leak), Fastlane, GitHub Actions & GitLab CI/CD (277 Tests 100% Pass).

- Architected AstroBite (v3.0.0) based on **Clean Architecture / Modular Package-based** structure with Riverpod 2.x code-gen across 12 decoupled business modules.
- Packaged proprietary **2D/3D Design System & Base Components** natively in Flutter (Custom Widgets, Theme Engine, Responsive Breakpoints), locking consistent **60 FPS**.
- Constructed **Base Network Client & Local Persistence**: Dio Interceptors (token refresh, retry policy), WebSocket & Firestore stream real-time, Hive/SQLite offline caching.
- Optimized **Render Tree & Memory via Flutter DevTools**: isolated painting via `RepaintBoundary`, `const` constructors, camera buffer disposal yielding sub-2.5s AI latency and **0 memory leaks**.
- Advanced **Web & Routing Optimization**: GoRouter & AutoRoute supporting nested routes, URL navigation, query params, CanvasKit/WASM rendering, and asset caching.
- Established a suite of **277 automated unit & widget tests (100% pass)**; standardized CI/CD pipelines (Web/Android/iOS) via Fastlane, GitHub Actions & GitLab CI/CD.

COMMERCIAL WORK EXPERIENCE

TRACK RECORD

FINVIET TECHNOLOGY

08/2025 – 08/2026

Mobile Developer (Flutter, Web & Kiosk) — Multi-Industry POS Kiosk Ecosystem (F&B, Retail & Spa Services)

Tech Stack: Flutter (Mobile, Web & Tablet), Dart, Clean Architecture, BLoC (flutter_bloc), Base Dio Client, SQLite/Drift, Platform Channels (Kotlin), ESC/POS Thermal, Fastlane, GitHub Actions.

- Architected modular **Clean Architecture / MVVM** applying **BLoC (flutter_bloc)** with Unidirectional Data Flow for multi-vertical POS & Kiosk checkouts (F&B, Retail, Spa), slashing transaction time by **30%**.
- Packaged **Base UI Library & Responsive Layout** dynamically adapting across screen form-factors (Mobile, Tablet, Kiosk/Web) with reusable Custom Widgets and dynamic Theme Engine.
- Built **Base Network Client with Dio**: configured Interceptors for automatic bearer token injection, 401 token refresh cycle, exponential backoff retry policy, and offline cache.
- Designed robust **Offline-first** data pipeline (BLoC, SQLite/Drift), ensuring 100% order persistence during network disconnects and auto-sync upon reconnection.
- Utilized **Flutter DevTools** (Profiler, Memory) to inspect Render Tree, eliminating drop frames/jank and avoiding RAM leaks when driving ESC/POS thermal printers and barcode scanners.
- Established **Coding conventions**, strict lint rules (`analysis_options.yaml`), led PR code reviews, and **mentored Junior developers** across the engineering team.
- Configured automated CI/CD build & release pipelines (Android APK/AAB, Web bundle) via **Fastlane & GitHub Actions / GitLab CI/CD**.

HAHALOLO SOCIAL TRAVEL NETWORK

05/2020 – 06/2025 (5 years)

Mobile & Frontend Developer (iOS & Flutter) — Ecosystem of 6 Apps: Social, OTT, Travel, Fintech & Outsource

Tech Stack: Flutter, Dart, Add-to-App, Clean Architecture, Riverpod, BLoC, iOS Swift, VIPER, MVVM, WebSocket (Starscream/Socket.IO), SQLite, CallKit, Firebase Crashlytics, GitLab CI/CD.

- **[Hahalolo — Travel Social Network (1M+ downloads)]**: Developed Tour, Flight, Hotel, Community modules; built Native iOS (Swift, VIPER) and integrated **Flutter via Add-to-App**; handled Responsive UI, Deep Linking, Crashlytics <0.5% crash rate. Mobile team: 20.
- **[Halome — OTT Realtime Messaging]**: Engineered real-time chat app from scratch using Native iOS Swift (MVVM); integrated **WebSocket real-time**, SQLite offline storage, RxSwift, CallKit VoIP; optimized for locked 60 FPS list scrolling. Team: 5.
- **[Haloki & Hahalolo Invest — Fintech]**: Maintained & optimized international money transfer app (192 countries) and stock trading platform in Flutter; ensured transaction integrity, KYC, real-time charts, and Dio interceptors.
- **[Aquannets — Outsource for Galactic Holdings]**: Developed aquaculture farm management app in Flutter applying **Clean Architecture + Riverpod**, SQLite offline-first at ponds. Team: 2.
- **[SocTrip — QC & Testing Lead]**: Led end-to-end (E2E) testing lifecycle, designed comprehensive test matrices, and provided quality gate sign-off for App Store & Google Play releases.

EDUCATION & CREDENTIALS

ACADEMIC

FPT POLYTECHNIC COLLEGE

Associate Degree in Information Technology (Software Development Major) • Proven practical competence across large-scale 1M+ user ecosystems

English: Professional Working (Technical documentation, architecture & collaboration) • **Vietnamese:** Native

ENGINEERING DISCIPLINE & METHODOLOGY

PRACTICES

Architecture & Performance: Clean Architecture (Package-based), Strict SOLID principles, Ponytail Mindset (Zero Bloat), Widget Lifecycle & Render Tree Optimization (DevTools 60 FPS, 0 memory leak).

Quality, Testing & Mentoring: 100% Deterministic Testing (Unit/Widget/Integration), Automated CI/CD (Fastlane, GitHub Actions, GitLab CI/CD, Docker), PR Code Review & Junior Mentoring.